Ch 3.1-2: (Multi)-Linear Regression

Lecture 5 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Sep 12, 2022

Announcements

Last time:

- Started 3.1 - Single linear regression

Announcements:

- Office Hours: Tues - Fri
- Homework #2 Due Weds, Sep 14
  - Upload solutions to CROWDMARK
  - Look for an email with the link to do this
  - Upload file for each question separately
- Quizzes
  - Approximately once a week
  - Look for email from crowdmark for feedback
  - Grades will be updated on D2L

Covered in this lecture

- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared
- Setup for multiple linear regression

Section 1

Last time

Setup

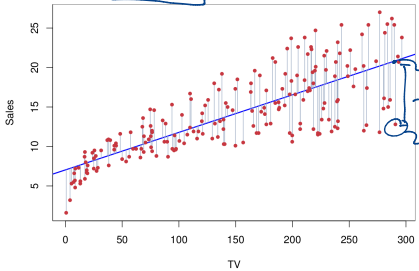
- Predict Y on a single predictor variable X

slope

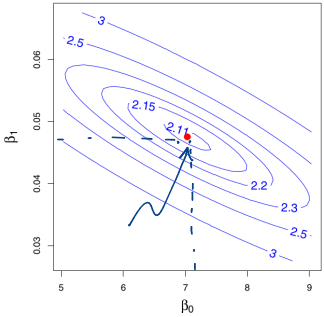
$$Y \approx \underbrace{\beta_0}_{\text{intercept}} + \beta_1 X$$

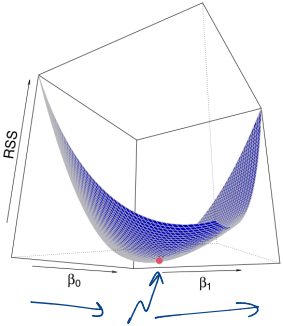
- "$\approx$" .... "is approximately modeled as"

- Given $(x_1, y_1), \cdots, (x_n, y_n)$ # data pts
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be prediction for Y on ith value of X.
- $e_i = y_i - \hat{y}_i$ is the ith residual



Least squares criterion: RSS





Residual sum of squares RSS is

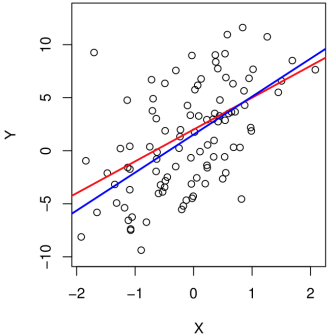
$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$
$$= \sum_i (y_i - \hat{y}_i)^2$$

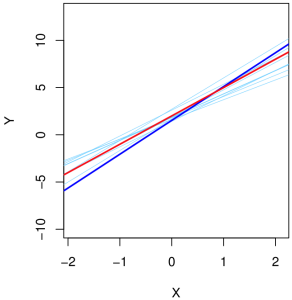
Least squares criterion

Find β_0 and β_1 that minimize the RSS.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Linear regression is unbiased





$\hat{\beta}_0$ $(\hat{\beta}_1)$ as a random variable is unbiased for β_0 (β_1)

Variance of linear regression estimates

Standard error

- Variance of linear regression estimates:

$$\mathrm{SE}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Residual standard error is an estimate of σ

$$RSE = \sqrt{RSS/(n-2)}$$

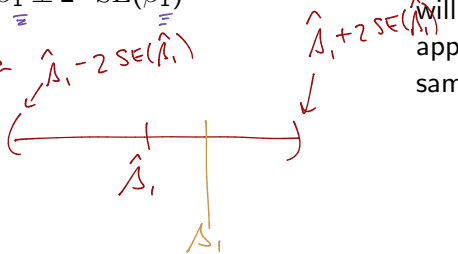
Confidence Interval

Same equation works for β_0

The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

Calc from data

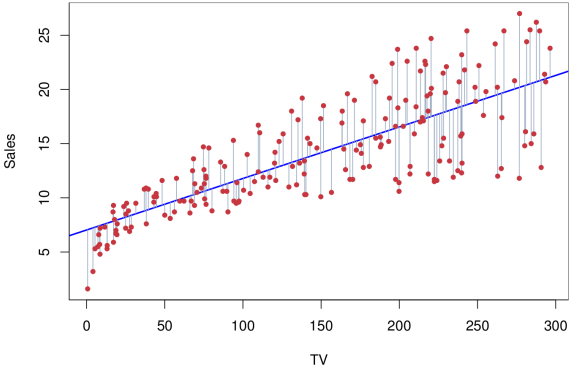


Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.

CI in Advertising data



For the advertising data set, the 95% CIs are:

- β_1 :: $[0.042, 0.053]$ } the slope is here
- β_0 :: $[6.130, 7.935]$

} intercept in here

Section 2

New stuff on evaluating models

Hypothesis testing

null hypothesis

$Y = \beta_0 + 0 \cdot X$

$Y = \beta_0$

H_0: There is no relationship between X and Y $\longrightarrow$ $H_0 : \beta_1 = 0$

H_a: There is some relationship between X and Y $\longrightarrow$ $H_a : \beta_1 \neq 0$

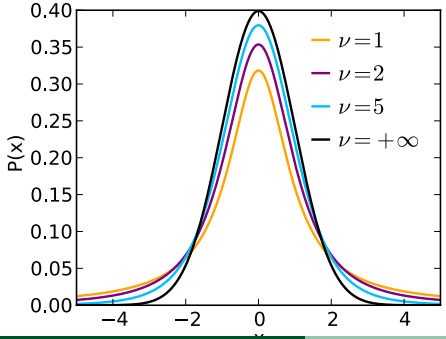
Test statistic and p-value

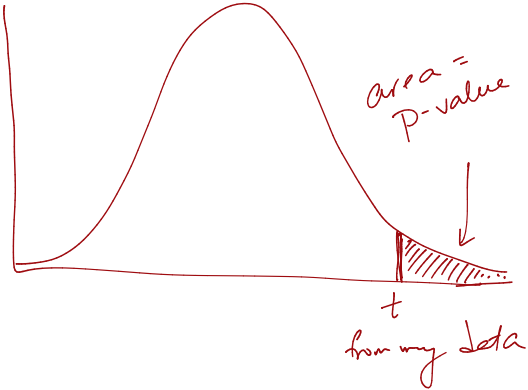
Test statistic:

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$$

gives a number based on data

t-distribution with $n - 2$ degrees of freedom

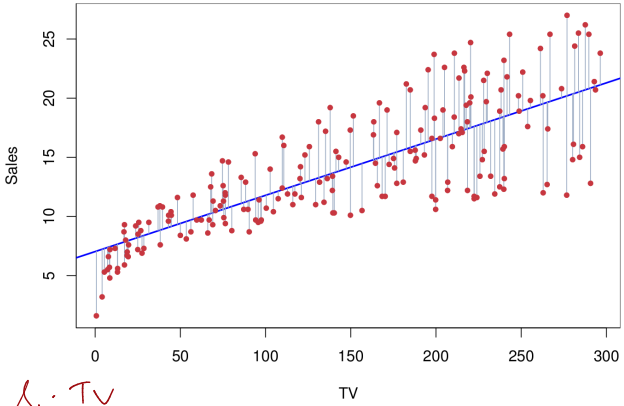




Advertising example

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Small p-value



- reject the null hypothesis
- there is some relationship between X and Y.

$\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV}$

Assessing the accuracy of the module: RSE

$Y = \beta_0 + \beta_1 X + \varepsilon$

Quantify the extent to which the model fits the data.

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$
$$= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}$$

- Estimate the std. dev. of ε
- Avg amt that the response will deviate from the true regression line

Assessing the accuracy of the module: R^2

$\sum_i (y_i - \hat{y}_i)^2$

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where total sum of squares is

$$TSS = \sum_i (y_i - \bar{y})^2$$

TSS = total variance in response without the regression

RSS = amt of variability after the regression

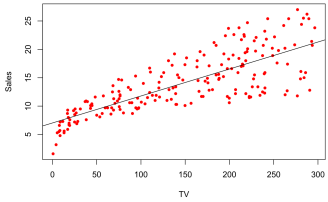
R^2 = prop of variability in Y that can be explained using X

Close to 0: Regression doesn't explain much variability

$0 \leq R^2 \leq 1$

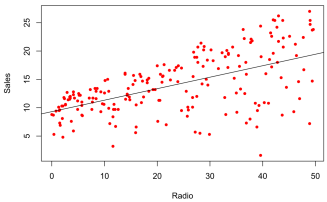
Close to 1: large prop of variance is explained by regression

Advertising example



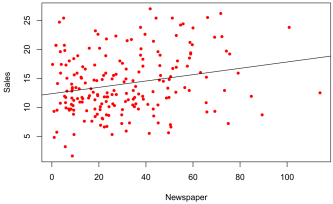
TV

$R^2 = 0.61$



Radio

$R^2 = 0.33$



Newspaper

$R^2 = 0.05$

Coding group work

Run the section titled "Assessing Coefficient Estimate Accuracy"

Section 3

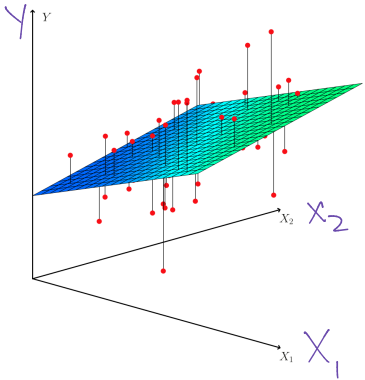
Multiple Linear Regression

Setup

hyper plane

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p + \varepsilon$$

β_j is the average effect on Y of one unit increase in X_j if all other X_i stay fixed



Estimation and Prediction

Given estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \cdots, \hat{\beta}_p$, prediction is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Minimize the sum of squares

$$\begin{aligned} RSS &= \sum_i (y_i - \hat{y}_i)^2 \\ &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \cdots - \hat{\beta}_p x_p) \end{aligned}$$

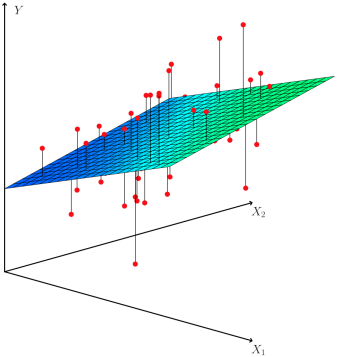
Coefficients are closed form but UGLY

Your code will do it for you

Advertising data set example

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

$= 2.939 + 0.046 \cdot TV + 0.189 \cdot radio - 0.001\, newspaper$



$\neq$

		Coefficient
	Intercept	2.939
X_1	TV	0.046
X_2	radio	0.189
X_3	newspaper	−0.001

Interpretation of coefficients

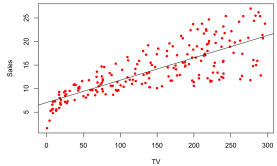
$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

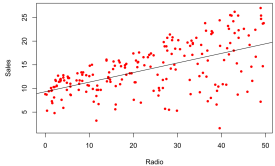
Fix TV + Newspaper spending

Spending \$1K more on radio results in 189 additional units of Sale

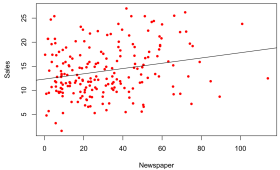
Single regression vs multi-regression



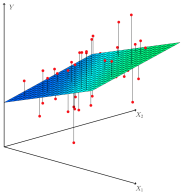
	Coefficient
Intercept	7.0325
TV	0.0475



	Coefficient
Intercept	9.312
radio	0.203



	Coefficient
Intercept	12.351
newspaper	0.055



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

Correlation matrix

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

In single lin reg:
newspaper is getting "credit" for radio.

Coding group work

Run the section titled "Multiple Linear Regression"

Next time

Lec #	Date		Topic	Reading	Homeworks
1	W	Aug 31	Intro / First day stuff / Python Review Pt 1	1	
2	F	Sep 2	What is statistical learning? / Python Review Pt 2	2.1	
	M	Sep 5	No class - Labor day		
3	W	Sep 7	Assessing Model Accuracy	2.2	HW #1 Due
4	F	Sep 9	Linear Regression	3.1	
5	M	Sep 12	More Linear Regression	3.2	
6	W	Sep 14	Even more linear regression	3.3	HW #2 Due
7	F	Sep 16	Probably more linear regression		
8	M	Sep 19	Intro to classification, Logisitic Regression	4.1, 4.2, 4.3	
9	W	Sep 21	More logistic regression		HW #3 Due
10	F	Sep 23	Review		
11	M	Sep 26	***Midterm #1***		
12	W	Sep 28	[No class, Dr Munch out of town]		
13	F	Sep 30	[No class, Dr Munch out of town]		
14	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
15	W	Oct 5	k-fold CV	5.1.3	
16	F	Oct 7	More k-fold CV	5.1.4	
17	M	Oct 10	CV for classification	5.1.5	HW #4 Due
18	W	Oct 12	Resampling methods: Bootstrap	5.2	
19	F	Oct 14	Subset selection	6.1	
20	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
21	W	Oct 19	Shrinkage: Lasso	6.2.2	
22	F	Oct 21	Dimension Reduction	6.3	